

Lab - 10

* Study of IP Addressing and sub-netting.

1. Find default subnet masks, network bits, host bits, hosts per subnet, no. of subnets, subnet numbers, 1st valid IP address, last valid IP address, and broadcast address.

(i) 8.1.4.5/16

(ii) 130.4.102.1/24

(iii) 130.4.102.1/22

(iv) 199.1.1.100/27

→ (i) 8.1.4.5/16

Class: A (1-126)

Default mask (class A): 255.0.0.0 (18)

Given mask: /16 $\Rightarrow$ 255.255.0.0

Network bits = 16, Host bits = 16

Hosts per subnet (usable) = $2^{16} - 2 = 65534$

Number of subnets borrowed bits = $16 - 8 = 8 \Rightarrow 2^8 = 256$ subnets

Subnet address: 8.1.0.0

1st valid host: 8.1.0.1, Last valid host: 8.1.255.254

Broadcast address: 8.1.255.255

(ii) 130.4.102.1/24

Class: B (128-191)

Default mask (class B): 255.255.0.0 (16)

Given mask: 124 $\Rightarrow$ 255.255.255.0

Network bits = 24, Host bits = 8

Hosts per subnet (usable) = $2^8 - 2 = 254$

Number of subnets borrowed = $24 - 16 = 8 \Rightarrow 2^8 = 256$ subnets

subnet address: 130.4.102.0

1st valid host: 130.4.102.1, last valid host: 130.4.102.254

Broadcast: 130.4.102.255

(iii) 130.4.102.1/22

class: B (128-191)

Default mask: 255.255.0.0 (/16)

Given mask: 122 $\Rightarrow$ 255.255.252.0

Network bits = 22, Host bits = 10

Hosts per subnet (usable) = $2^{10} - 2 = 1022$

Number of subnets borrowed = $22 - 16 = 6 \Rightarrow 2^6 = 64$

subnet block size in 3rd octet = $256 - 252 = 4$ subnets

Network address: 130.4.100.0

1st valid host: 130.4.100.1, last valid host: 130.4.103.254

Broadcast: 130.4.103.255

(iv) 199.1.100/27

class: B (192-223)

Default mask: 255.255.255.0 (/24)

Given mask: 127 $\Rightarrow$ 255.255.255.0

Network bits = 27, Host bits = 5

Host bits = 27, Host bits = 5

Host per subnet (usable) = $2^5 - 2 = 30$

Number of subnet borrowed = $27 - 24 = 3 \Rightarrow 2^3 = 8$ subnets

Block size in last octet = $256 - 224 = 32$

subnet : floor $(100/32) = 3$, Network address = $199.1.196$

1st Valid host : $199.1.197$, last valid host : $199.1.1.226$

Broadcast : $199.1.1.227$

2. A host in a class C network has been assigned an IP address $192.168.17.9$. Find the number of addresses in the block, the first address, and the last address.

→ Default mask : $255.255.255.0$ (124)

Number of addresses in block = $2^8 = 256$

First address : $192.168.17.0$, Last address : $192.168.17.255$

3. An address in a block is given as $185.28.17.9$. Find the number of addresses in the block, the first address, and the last address.

→ Number of address in block = $2^{16} = 65536$

First Address : $185.28.0.0$, last address = $185.28.255.255$

4. A block of addresses is granted to a small Organization. We know that one of the addresses is $205.16.37.39/28$. What is the first address, last address, number of address in a block?

→ First address : $205.16.37.32$, Last Address : $205.16.37.47$

Address Number of address in block = $2^{32-28} = 16$

5. subnet the IP address 216.25.5.0 into 30 hosts in each subnet. Find class, Default mask, subnet mask, bit borrowed, new subnet mask, no. of hosts and subnet, network Ranges (Subnets).

→ Class: C

Default ^{new} mask: $32 - 5 = 27 \Rightarrow 255.255.255.224$

bit borrowed: $8 - 5 = 3$

Default mask: 255.255.255.0 (124)

no. of hosts: $2^5 - 2 = 30$

No. of subnets = $2^3 = 8$

1. 216.21.5.0/27 - 216.21.5.31/27

2. 216.21.5.32/27 - 216.21.5.63/27

3. 216.21.5.64/27 - 216.21.5.95/27

4. 216.21.5.96/27 - 216.21.5.127/27

5. 216.21.5.128/27 - 216.21.5.159/27

6. 216.21.5.160/27 - 216.21.5.191/27

7. 216.21.5.192/27 - 216.21.5.223/27

8. 216.21.5.224/27 - 216.21.5.255/27

6. Subnet the IP address 192.10.20.0 into 52 hosts in each subnet. Find class, Default mask, bit borrowed, New subnet mask, No. of hosts and subnet, network Ranges (Subnets).

→ Class: C

Default mask = 124 $\Rightarrow 255.255.255.0$

bits borrowed: $8 - 6 = 2$

new subnet mask: 126 $\Rightarrow 255.255.255.192$

Hosts per subnet = $2^6 = 64$

No. of subnets = $2^2 = 4$

Block size = 64

1. 192.10.20.0/26 - 192.10.20.63/26

2. 192.10.20.64/26 - 192.10.20.127/26

3. 192.10.20.128/26 - 192.10.20.191/26

4. 192.10.20.192/26 - 192.10.20.255/26

7. Determining the subnet mask for Devices A and B:

(a) Device A: 172.16.17.30/20

(b) Device B: 172.16.28.15/20

→ (a) 172.16.17.30/20

mask: 20/20 $\Rightarrow$ 255.255.240.0

Block size: $256 - 240 = 16$

Network address = 172.16.16.0/20

(b) 172.16.28.15/20

mask: 20 $\Rightarrow$ 255.255.240.0

Block size: $256 - 240 = 16$

Network address = 172.16.16.0/20